

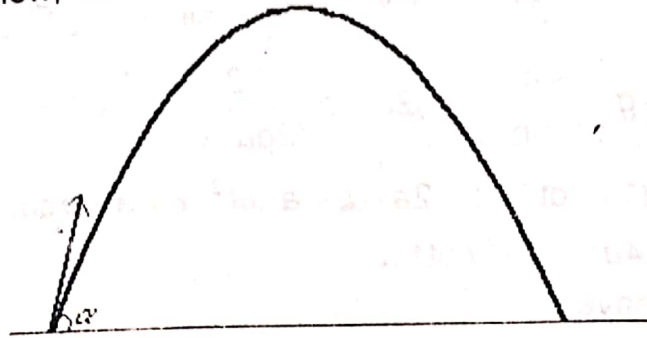
1. Tick ✓ the correct option in the following questions:

- (i) If a particle is projected making a certain angle with horizontal into the air with a certain velocity or dropped from a moving plane then it is called a
(a) ✓ projectile (b) hyperbola (c) circle (d) straight line
- (ii) The path of a projectile is called its
(a) velocity (b) acceleration (c) ✓ trajectory (d) none of these
- (iii) Angle with which the particle is projected is called the _____ of projection.
(a) velocity (b) acceleration (c) ✓ angle (d) none of these
- (iv) The velocity with which the particle is projected is called the _____ of projection.
(a) ✓ velocity (b) acceleration (c) angle (d) none of these

If we neglect the air resistance, a projectile is a particle moving under the influence of _____ force.

- (a) frictional (b) ☒ gravitational (c) angle (d) none of these

In the figure below, the horizontal component of velocity of projection is



(a) ☒ $v_0 \cos \alpha$

(b) $v_0 \sin \alpha$

(c) $v_0 \tan \alpha$

(d) $v_0 \cot \alpha$

i) In the figure below,



the vertical component of velocity of projection is

(a) $v_0 \cos \alpha$

(b) ☒ $v_0 \sin \alpha$

(c) $v_0 \tan \alpha$

(d) $v_0 \cot \alpha$

ii) In the figure below,



the horizontal component of velocity of projection is

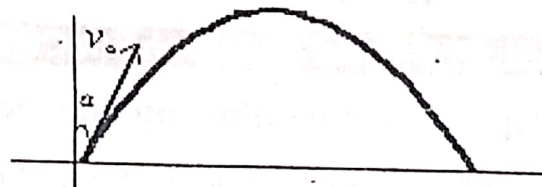
(a) $v_0 \cos \alpha$

(b) ☒ $v_0 \sin \alpha$

(c) $v_0 \tan \alpha$

(d) $v_0 \cot \alpha$

In the figure below,



the vertical component of velocity of projection is

(a) ☒ $v_0 \cos \alpha$

(b) $v_0 \sin \alpha$

(c) $v_0 \tan \alpha$

(d) $v_0 \cot \alpha$

The equation of path of projectile is $y =$

(a) $x \sin \alpha - \frac{gx^2}{2v_0^2} \sec^2 \alpha$

(b) ☒ $x \tan \alpha - \frac{gx^2}{2v_0^2} \sec^2 \alpha$

(c) $x \tan \alpha - \frac{gx^2}{2v_0^2} \cos^2 \alpha$

(d) none of these

- xi) In the equation of path of projectile $y = x \tan \alpha - \frac{gx^2}{2v_o^2} \sec^2 \alpha$, v_o is the
 (a) final velocity (b) terminal velocity
~~(c) velocity of projection~~ (d) none of these
- xii) If 45° and v_o are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$
~~(a) $x - \frac{gx^2}{v_o^2}$~~ (b) $\sqrt{3}x - \frac{2gx^2}{v_o^2}$ (c) $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_o^2}$ (d) $x - gv_o^2 x^2$
- xiii) If 60° and v_o are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$
 (a) $x - \frac{gx^2}{v_o^2}$ ~~(b) $\sqrt{3}x - \frac{2gx^2}{v_o^2}$~~ (c) $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_o^2}$ (d) $x - gv_o^2 x^2$
- xiv) If 30° and v_o are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$
 (a) $x - \frac{gx^2}{v_o^2}$ (b) $\sqrt{3}x - \frac{2gx^2}{v_o^2}$ ~~(c) $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_o^2}$~~ (d) $x - gv_o^2 x^2$
- xv) If 45° and $\frac{1}{3}$ m/s are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$
 (a) $x - \frac{gx^2}{v_o^2}$ (b) $\sqrt{3}x - \frac{2gx^2}{v_o^2}$ (c) $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_o^2}$ ~~(d) $x - 9gx^2$~~
- xvi) The path of a projectile is a
 (a) straight line (b) circle ~~(c) parabola~~ (d) none of these
- xvii) The vertex of the path of projectile is
~~(a) $\left(\frac{v_o^2 \sin 2\alpha}{2g}, \frac{v_o^2 \sin^2 \alpha}{2g}\right)$~~ (b) $\left(\frac{v_o^2 \cos 2\alpha}{2g}, \frac{v_o^2 \cos^2 \alpha}{2g}\right)$
 (c) $\left(\frac{v_o^2 \sec 2\alpha}{2g}, \frac{v_o^2 \sec^2 \alpha}{2g}\right)$ (d) (0,0)
- xviii) If 30° and v_o are the angle and velocity of projection respectively, then the vertex of the path of projectile is
 (a) $\left(\frac{2\sqrt{3}v_o^2}{g}, \frac{v_o^2}{g}\right)$ (b) $\left(\frac{\sqrt{3}v_o^2}{g}, \frac{v_o^2}{2g}\right)$ (c) $\left(\frac{\sqrt{3}v_o^2}{2g}, \frac{v_o^2}{4g}\right)$ ~~(d) $\left(\frac{\sqrt{3}v_o^2}{4g}, \frac{v_o^2}{8g}\right)$~~
- xix) If 45° and v_o are the angle and velocity of projection respectively, then the vertex of the path of projectile is

- (a) $\left(\frac{2\sqrt{3}v_o^2}{g}, \frac{v_o^2}{g}\right)$ (b) $\left(\frac{v_o^2}{2g}, \frac{v_o^2}{4g}\right)$ (c) $\left(\frac{\sqrt{3}v_o^2}{4g}, \frac{3v_o^2}{8g}\right)$ (d) $\left(\frac{\sqrt{3}v_o^2}{4g}, \frac{v_o^2}{8g}\right)$
- (xx) If 60° and v_o are the angle and velocity of projection respectively, then the vertex of the path of projectile is

- (a) $\left(\frac{2\sqrt{3}v_o^2}{g}, \frac{v_o^2}{g}\right)$ (b) $\left(\frac{v_o^2}{2g}, \frac{v_o^2}{4g}\right)$ (c) $\left(\frac{\sqrt{3}v_o^2}{4g}, \frac{3v_o^2}{8g}\right)$ (d) $\left(\frac{\sqrt{3}v_o^2}{4g}, \frac{v_o^2}{8g}\right)$

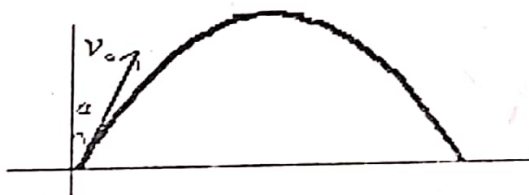
2. Tick ✓ the correct option in the following questions:

- (i) The latus rectum of the path of projectile is
 (a) $\frac{2v_o^2 \sin^2 \alpha}{g}$ (b) $\frac{2v_o^2 \cos^2 \alpha}{g}$ (c) $\frac{v_o^2}{2g}$ (d) $2v_o^2 \sin^2 \alpha$
- (ii) The directrix of the path of projectile is
 (a) $2v_o^2 \tan^2 \alpha$ (b) $2v_o^2 \cos^2 \alpha$ (c) $2v_o^2 \sin^2 \alpha$ (d) $\frac{v_o^2}{2g}$
- (iii) The distance covered along horizontal direction by the projectile from point projection to the point where the projectile strikes the ground is called the
 (a) horizontal range (b) vertical range
 (c) horizontal velocity (d) none of these
- (iv) The horizontal range of a projectile is $R =$
 (a) $\frac{v_o^2 \tan 2\alpha}{g}$ (b) $\frac{v_o^2 \sin 2\alpha}{g}$ (c) $\frac{v_o^2 \cos 2\alpha}{g}$ (d) $v_o^2 \sin 2\alpha$
- (v) If 15° and v_o are the angle and velocity of projection, then the horizontal range of projectile is
 (a) $\frac{v_o^2}{2g}$ (b) $\frac{\sqrt{3}v_o^2}{2g}$ (c) $\frac{v_o^2}{g}$ (d) $v_o^2 g$
- (vi) If 30° and v_o are the angle and velocity of projection, then the horizontal range of projectile is
 (a) $\frac{v_o^2}{2g}$ (b) $\frac{\sqrt{3}v_o^2}{2g}$ (c) $\frac{v_o^2}{g}$ (d) $v_o^2 g$
- (vii) The horizontal range of a projectile is maximum if angle of projection is
 (a) 15° (b) 30° (c) 45° (d) 60°
- (viii) The maximum horizontal range of a projectile is
 (a) $\frac{v_o^2}{2g}$ (b) $\frac{\sqrt{3}v_o^2}{2g}$ (c) $\frac{v_o^2}{g}$ (d) $v_o^2 g$
- (ix) If 7 ft/s is the velocity of projection, then the maximum horizontal range is
 (a) $\frac{49}{32}$ ft (b) $\frac{9}{8}$ ft (c) $\frac{32}{49}$ ft (d) none of these

- (x) If 8 ft/s is the velocity of projection, then the maximum horizontal range is
 (a) 6 ft (b) 4 ft (c) 3 ft ☒ (d) 2 ft
- (xi) If 80 ft/s is the velocity of projection, then the maximum horizontal range is
 (a) 10 ft (b) 100 ft ☒ (c) 200 ft (d) 250 ft
- (xii) The horizontal range of the projectile shown in the figure below is



- ☒ (a) $\frac{v_0^2 \sin 2\alpha}{g}$ (b) $\frac{v_0^2 \cos 2\alpha}{g}$ (c) $\frac{v_0^2 \tan 2\alpha}{g}$ (d) $v_0^2 g \sin 2\alpha t$
- (xiii) The maximum horizontal range of a projectile is 100 ft, the velocity of projection is
 (a) 3.6 ft/s (b) 4.6 ft/s (c) 5.6 ft/s ☒ (d) 56.57 ft/s
- (xiv) The time of flight of a projectile is $t =$
 (a) $\frac{v_0 \sin \alpha}{2g}$ (b) $\frac{v_0 \sin \alpha}{g}$ ☒ (c) $\frac{2v_0 \sin \alpha}{g}$ (d) $2v_0 g \sin \alpha$
- (xv) The time taken to reach the maximum height of a projectile is
 (a) $\frac{v_0 \sin \alpha}{2g}$ ☒ (b) $\frac{v_0 \sin \alpha}{g}$ (c) $\frac{2v_0 \sin \alpha}{g}$ (d) $2v_0 g \sin \alpha$
- (xvi) The time of flight of the projectile shown in the figure below is



- (a) $\frac{v_0 \sin \alpha}{2g}$ (b) $\frac{v_0 \sin \alpha}{g}$ (c) $\frac{2v_0 \sin \alpha}{g}$ ☒ (d) $\frac{2v_0 \cos \alpha}{g}$
- (xvii) The time taken by the projectile to attain the maximum height of the projectile in the figure below is



- (a) $\frac{v_0 \sin \alpha}{2g}$ (b) $\frac{v_0 \sin \alpha}{g}$ ☒ (c) $\frac{v_0 \cos \alpha}{g}$ (d) $\frac{2v_0 \cos \alpha}{g}$
- (xviii) If 45° and v_0 are the angle and velocity of projection respectively, then the time of flight of the projectile is

- ☒ (a) $\frac{\sqrt{2} v_0}{g}$ (b) $\frac{v_0}{g}$ (c) $\frac{\sqrt{3} v_0}{g}$ (d) $\frac{v_0}{\sqrt{2} g}$

(xix) If 30° and v_0 are the angle and velocity of projection respectively, then the time of flight of the projectile is

- (a) $\frac{\sqrt{2}v_0}{g}$ (b) $\frac{v_0}{g}$ (c) $\frac{\sqrt{3}v_0}{g}$ (d) $\frac{v_0}{\sqrt{2}g}$

(xx) If 60° and v_0 are the angle and velocity of projection respectively, then the time of flight of the projectile is

- (a) $\frac{\sqrt{2}v_0}{g}$ (b) $\frac{v_0}{g}$ (c) $\frac{\sqrt{3}v_0}{g}$ (d) $\frac{v_0}{\sqrt{2}g}$

Tick ✓ the correct option in the following questions:

i) If 45° and v_0 are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is

- (a) $\frac{\sqrt{2}v_0}{g}$ (b) $\frac{v_0}{g}$ (c) $\frac{\sqrt{3}v_0}{g}$ (d) $\frac{v_0}{\sqrt{2}g}$

ii) If 30° and v_0 are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is

- (a) $\frac{v_0}{2g}$ (b) $\frac{\sqrt{3}v_0}{2g}$ (c) $\frac{v_0}{g}$ (d) none of these

iii) If 60° and v_0 are the angle and velocity of projection respectively, then the time of flight of the projectile is

- (a) $\frac{v_0}{2g}$ (b) $\frac{\sqrt{3}v_0}{2g}$ (c) $\frac{v_0}{g}$ (d) none of these

iv) A projectile is projected with speed 30 m/s at an angle of 30° to the horizontal, the time of flight is

- (a) 4 s (b) 3 s (c) 2 s (d) 1 s

The maximum height attained by the projectile is

- (a) $\frac{v_0^2 \sec^2 \alpha}{2g}$ (b) $\frac{v_0^2 \tan^2 \alpha}{2g}$ (c) $\frac{v_0^2 \cos^2 \alpha}{2g}$ (d) $\frac{v_0^2 \sin^2 \alpha}{2g}$

The speed at a point P of the projectile is _____ as attained by a particle of the same mass at P falling freely from the point vertically above P on the directrix of the trajectory.

- (a) less than that (b) same (c) greater than that (d) none of these

The least speed to hit the projectile a point (a, b) is

- (a) $\sqrt{g(b + \sqrt{a^2 + b^2})}$ (b) $g(b + \sqrt{a^2 + b^2})$

- (c) $[g(b + \sqrt{a^2 + b^2})]^2$ (d) none of these

The least speed to hit the projectile a point $(1, 2)$ is

- (ix) The least speed to hit the projectile a point (3,4) is
 (a) $g(2 + \sqrt{5})$ (b) $\sqrt{g(2 + \sqrt{5})}$ (c) $3\sqrt{g}$ (d) $3g$
- (x) The range on an inclined plane of inclination β is
 (a) $\frac{2v_o^2 \cos \alpha \sin(\alpha - \beta)}{g \sin^2 \beta}$ (b) $\frac{2v_o^2 \cos \alpha \sin(\alpha - \beta)}{g \cos^2 \beta}$ (c) $\frac{2v_o^2 \cos \alpha \sin(\alpha - \beta)}{g \tan^2 \beta}$ (d) none of these
- (xi) The maximum range on an inclined plane of inclination β is
 (a) $\frac{v_o^2}{g(1 + \sin \beta)}$ (b) $\frac{v_o^2}{2g(1 + \sin \beta)}$ (c) $\frac{v_o^2}{3g(1 + \sin \beta)}$ (d) none of these
- (xii) The maximum range on an inclined plane of inclination $\beta = 30^\circ$ is
 (a) $\frac{v_o^2}{3g}$ (b) $\frac{2v_o^2}{3g}$ (c) $\frac{v_o^2}{g}$ (d) gv_o^2
- (xiii) The maximum range down the inclined plane of inclination $-\beta$ is
 (a) $\frac{v_o^2}{g(1 + \sin \beta)}$ (b) $\frac{v_o^2}{2g(1 + \sin \beta)}$ (c) $\frac{v_o^2}{3g(1 + \sin \beta)}$ (d) $\frac{v_o^2}{g(1 - \sin \beta)}$
- (xiv) The time of flight on an inclined plane of inclination β is $t =$
 (a) $\frac{2v_o \cos(\alpha + \beta)}{g \cos \beta}$ (b) $\frac{2v_o \sin(\alpha + \beta)}{g \cos \beta}$ (c) $\frac{2v_o \sin(\alpha - \beta)}{g \cos \beta}$ (d) none of these
- (xv) Time of flight down the inclined plane of inclination $-\beta$ is $t =$
 (a) $\frac{2v_o \cos(\alpha + \beta)}{g \cos \beta}$ (b) $\frac{2v_o \sin(\alpha + \beta)}{g \cos \beta}$ (c) $\frac{2v_o \sin(\alpha - \beta)}{g \cos \beta}$ (d) none of these
- (xvi) The locus of all points on different trajectories which lie at maximum distances from point of projection is called _____ of safety.
 (a) hyperbola (b) ellipse (c) parabola (d) circle
- (xvii) The equation of parabola of safety is $x^2 =$
 (a) $-\frac{2v_o^2}{g} \left(y - \frac{2v_o^2}{g} \right)$ (b) $-\frac{2v_o^2}{g} \left(y - \frac{v_o^2}{g} \right)$ (c) $-\frac{2v_o^2}{g} \left(y - \frac{v_o^2}{2g} \right)$ (d) none of these
- (xviii) The directrix of the parabola of safety is
 (a) $\frac{v_o^2}{4g}$ (b) $\frac{v_o^2}{2g}$ (c) $\frac{v_o^2}{g}$ (d) $\frac{2v_o^2}{g}$
- (xix) The vertex of the parabola of safety is

(a) $\left(0, \frac{4v_0^2}{g}\right)$

(b) $\left(0, \frac{2v_0^2}{g}\right)$

(c) $\left(0, \frac{v_0^2}{g}\right)$

☒ (d) $\left(0, \frac{v_0^2}{2g}\right)$

The focus of the parabola of safety is

(a) (1,2)

(b) (2,1)

(c) (1,1)

☒ (d) (0,0)

Tick ✓ the correct option in the following questions:

A particle projected with speed v_0 and angle of projection α have the maximum range as:

(a) $\frac{v_0^2 \sin 2\alpha}{g}$

(b) $\frac{v_0^2 \cos 2\alpha}{g}$

☒ (c) $\frac{v_0^2}{g}$

(d) $\frac{v_0}{g^2}$

PU, 2015 (BS Math)

The motion of a particle in air with velocity v_0 in the direction making an angle α with the horizontal is

☒ (a) parabola

(b) circle

(c) ellipse

(d) straight line

PU, 2017 (BS Math)

Parabola of safety is also called:

☒ (a) Fixed parabola

(c) Circular parabola

(b) Rectangular parabola

(d) Harmonic parabola

PU, 2015 (BS Math)